
Earning the Congress's Vote: An Eight-Question Passage Framework for Enacting H. R. 9510 in the House and Senate

Kevin Kawchak ^{ID}
CEO ChemicalQDevice
10.5281/zenodo.xxxxxxxx

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Abstract.

H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026, would amend the Federal Food, Drug, and Cosmetic Act to require that an automated verification, validation, and uncertainty quantification examination clear robot-patient interaction code before that code is generated or executed in an oncology clinical investigation. This paper is a passage framework: it organizes the case for the bill as the eight questions a member of Congress asks before a yes vote, and pairs each question with a cited fact drawn from the bill text, the National Platform that demonstrates the bill is implementable, and the public record. The eight questions are the mandate for a statute, Congress's authority to act, the safety the bill creates, the fiscal score, the benefit to constituents, bipartisanship, the supporting coalition, and the passage path itself. The answers are concrete: a ten-gate safety floor bound to published consensus standards; an authorization of 58 million dollars over five years with no new mandatory spending and a verification cost roughly nineteen times lower than conventional review; and a platform that, in validated simulation, treated 168 patients with 29 robots at 99.7 percent uptime with zero patient harm events. The framework is written for the staffer, the member, and the committee counsel who must decide whether to mark up, cosponsor, and vote for the bill in both chambers. It does not rewrite the prior narrative review or the clinician confidence framework; it is a new, analogous companion with one objective: making H. R. 9510 most likely to pass in both the House and the Senate.

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Keywords H. R. 9510 | Verification before generation | Physical AI oncology trials | Legislative passage | VVUQ | Budget score | Bipartisanship | FD&C Act section 515D

1 Introduction: the eight questions behind a yes vote

A bill does not pass because it is correct. It passes because enough members of each chamber can answer, to their own satisfaction and to a constituent’s, the small set of questions that stand between a draft and a statute. This paper sets out those questions for H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026, and answers each with evidence the member can cite [1].

The subject is unfamiliar to most members, so the terms are defined before they are used. Physical AI is the integration of embodied robots with advanced artificial intelligence for direct patient care in a clinical trial [2]. Robot-patient interaction code is the machine-generated software that directs a robot’s behavior at the moment it touches a patient. Verification, validation, and uncertainty quantification, abbreviated VVUQ, is an examination that asks whether the action was built correctly, whether it is the right action for this patient, and how confident the system is. Verification before generation is the bill’s central rule: the examination must clear before the code is written or run, not after. The examination is a suite of ten gates, each bound to a published safety standard, and each action it reviews is resolved to one of three outcomes, ACCEPT and act under audit, ESCALATE to a qualified human, or BLOCK before the patient. The bill writes this rule into a new section 515D of the Federal Food, Drug, and Cosmetic Act. Table 1 collects these and the budget terms a member needs.

Term	Plain-language meaning for a legislator
Physical AI	Embodied robots guided by advanced AI for direct patient care in a trial.
Robot-patient interaction code	The software that directs a robot at the moment it contacts a patient.
VVUQ	An automated check of whether an action is built right, is right for the patient, and how certain the system is.
Verification before generation	The rule that the check must clear before code is generated or executed.
Ten gates; ACCEPT, ESCALATE, BLOCK	The ten-part check and its three outcomes; three gates are hard catastrophe predicates that must always hold.
Section 515D	The new section of the FD&C Act the bill adds (21 U.S.C. 360e-5).
Authorization versus appropriation	Authorization sets a ceiling Congress may later fund; an appropriation is the actual money. This bill only authorizes.
PAYGO and CutGo	Budget rules requiring that new mandatory spending be offset; this bill creates none.

Table 1: The terms a member needs before reading the framework, defined before first substantive use.

The argument is an ordered path. The eight questions run from whether a statute is needed at all, through authority, safety, and the budget score, to the people the bill helps, the politics of the two caucuses, the coalition outside the chamber, and finally the path through both chambers. ?? traces that path; each later section answers one question and closes on the specific provision, score line, or evidence that settles it.

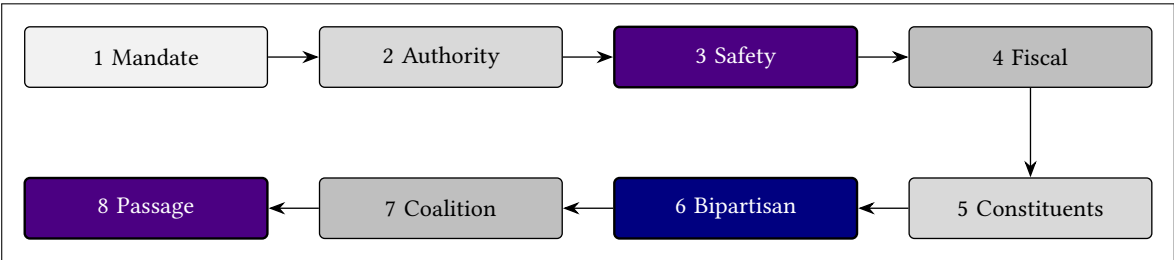


Figure 1. The eight passage questions as one ordered path, from the mandate for a statute to the path through both chambers; the colored TikZ reproduction of catalog diagram 02.

The objective of the framework is not to win an argument. It is to assemble a majority in the House and the votes to pass the Senate, by giving each member an answer they can defend at home and in a markup.

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2 Mandate: is the problem real enough to need a Federal statute?

The first question is whether there is a problem only Congress can solve. A member will not spend political capital on a bill that duplicates existing law or chases a hypothetical harm, so the mandate has to be specific. H. R. 9510 rests on three gaps in current law, set out in the bill's findings, that no amount of agency guidance has closed [1].

The first is the embodiment problem and the sequencing gap. Physical AI oncology systems place autonomous and semi-autonomous robots in direct contact with patients, and an error in the machine-generated code that directs a robot cannot be caught after the fact the way a paperwork error can; by the time the action executes, the patient has already been touched. No Federal law regulates the order of operations, so nothing today requires that verification happen before the code is generated. The second is the financial-data gap. Existing law already makes a clinical investigator's financial arrangements reportable through the Open Payments system [3], but no law prices, records, or discloses the cost of verifying robot-patient interaction code, so the one number a budget committee would want does not exist. The third is the cost asymmetry: an embodied failure is priced in lives and litigation, while the verification gate that prevents it is priced in compute, storage, and a bounded amount of human review. The gate's cost is small, fixed, and measurable; the failure it prevents is large and unbounded.

The human stakes are not abstract. The foundational estimate that tens of thousands of Americans die each year from preventable medical error has anchored two decades of safety legislation [4], and later analysis placed medical error among the leading causes of death in the United States [5]. A statute that makes a specific, automatable safety check mandatory before an autonomous robot acts is a direct response to that record, not a speculative one. Table 2 states each gap and what it leaves unaddressed today.

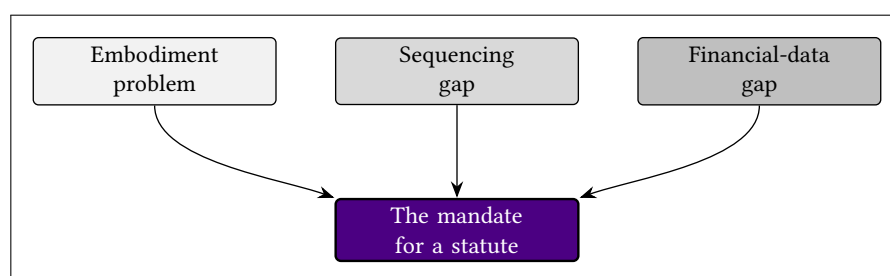


Figure 2. Three gaps in existing law compose the mandate: the embodiment and sequencing gap, the financial-data gap, and the cost asymmetry that makes the gate cheap relative to the harm it prevents.

Gap in existing law	What it leaves unaddressed today
Embodiment and sequencing	An error in robot-patient interaction code cannot be caught in time once a robot is in contact with a patient, and no Federal law requires verification before the code is generated or executed.
Financial-data	No Federal law prices, records, or discloses the cost of verifying robot-patient interaction code, so no comparable number reaches a budget committee.
Cost asymmetry	Nothing in current law forces the small, fixed cost of a verification gate to be weighed against the large, unbounded cost of an embodied failure.

Table 2: The mandate: the gaps in existing law that only a Federal statute closes.

The mandate question is answered not by alarm but by specificity: three named gaps, each closed by a single new section, each tied to a harm a member can explain to a constituent who lost someone to a preventable error.

3 Authority: is this within Congress's power and consistent with existing law?

The second question is whether Congress may act here at all, and whether the bill respects the law already on the books. A clean answer disarms both the constitutional objection and the federalism objection before they are raised.

Congress’s authority is the same it has exercised over medical devices for decades. H. R. 9510 amends the Federal Food, Drug, and Cosmetic Act, 21 U.S.C. 301 et seq., an exercise of the commerce power, and adds a single new section 515D (21 U.S.C. 360e-5) together with ten conforming amendments that thread the new requirement through the existing device provisions, from the definition of a device in section 321(h) to the State preemption rule in section 360k [6, 1]. The bill does not invent a new regulatory regime; it places a sequencing requirement inside the one Congress already built and the FDA already administers.

The bill is careful with existing law. Its rule of construction states that nothing in section 515D displaces the investigational-device exemption of section 520(g), the human-subjects protections of 21 CFR part 50, or the investigational new drug framework of 21 CFR part 312, and that nothing reclassifies any device. Those are the same frameworks the National Platform adapts rather than replaces [7, 8, 9]. The closest existing analogues confirm that the bill fills a gap rather than overlapping: section 515C predetermined change control plans and the FDA’s guidance on AI-enabled device software are the nearest precedents, but both are voluntary and neither requires verification before generation, while the strongest Federal precedent for human oversight of an algorithm, the Medicare Advantage rule on the use of algorithms in coverage determinations, governs payment rather than a robot at the bedside [10].

Federalism is handled by a savings clause, not by silence. The new section 360k(c) preserves State requirements that a licensed human retain authority to monitor, override, or terminate an AI or robotic system, and that audit records be kept, so long as those requirements are not different from or in addition to the Federal floor. ?? shows authority flowing from the commerce power through the FD&C Act to a Federal floor that leaves State human-over-AI law standing.

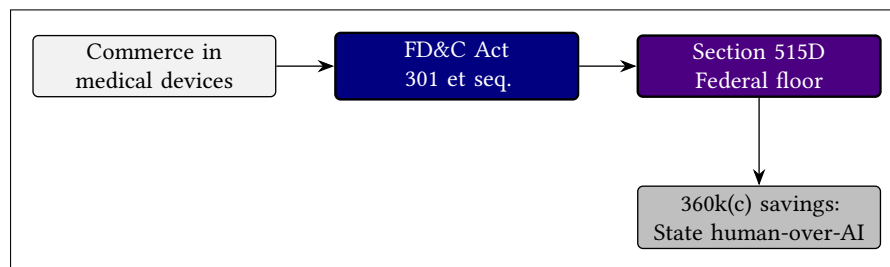


Figure 3. Authority flows from the commerce power through the FD&C Act to a Federal floor in section 515D that preserves State human-over-AI law; the colored TikZ reproduction of catalog diagram 09.

Rule-of-construction clause	The existing law it preserves
Investigational-device exemption	Section 520(g) of the FD&C Act is undisturbed.
Human-subjects protection	21 CFR part 50 continues to govern consent and protection.
Investigational new drug framework	21 CFR part 312 continues to govern the trial.
Change control outside Physical AI	Section 515C authority is unchanged for other devices.
State human-over-AI requirements	New section 360k(c) preserves them as a floor, not a ceiling.

Table 3: Authority with restraint: each rule-of-construction clause names the existing law the bill leaves in place.

Authority is answered by precedent and restraint: a familiar commerce-power amendment to a statute the FDA already runs, with an explicit promise about what it does not touch.

4 Safety: what in the bill actually protects patients?

The third question is the one a member will be asked on local television: what, in plain terms, keeps a robot from hurting someone? The answer must be concrete, and it must be in the statute rather than in a press release.

The mechanism is verification before generation. Before any robot-patient interaction code is generated or executed, the system's proposed action is examined by a ten-gate VVUQ suite, and the action is resolved to ACCEPT and proceed under audit, ESCALATE to a qualified human, or BLOCK before it can reach the patient. The decision rule in section 515D is strict: a single failing dimension forces a BLOCK, any ambiguity or divergence forces an ESCALATE, and the verification fraction, the share of interactions that must clear every gate, is fixed at 1.0 [1]. ?? shows the pipeline.

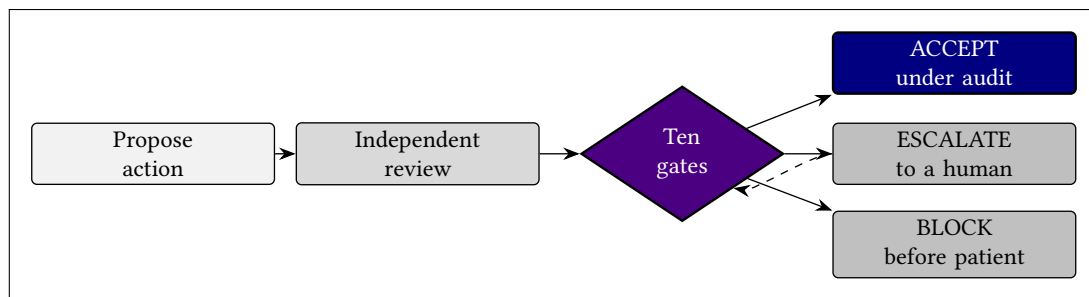


Figure 4. Verification before generation: every proposed action is gated to ACCEPT, ESCALATE, or BLOCK before the patient; the colored TikZ reproduction of catalog diagram 03.

What makes the floor credible is that the bill does not invent its own rigor. Each of the ten gates is bound to one or more published consensus standards, with a minimum validation agreement and a maximum coefficient of variation, and three of the gates, the vascular no-fly volume, the shared-room human collaboration, and the fault and emergency-stop behavior, are hard catastrophe predicates that must hold at an agreement of 1.00 without exception [11, 12, 13, 14, 15]. Table 4 reproduces the schedule. That the suite is not theoretical is shown by the author's assurance lineage, in which a surgical humanoid was cleared only after the same gate discipline was applied [16, 17].

#	Gate	Agree.	CV	Bound standards
1	Bimanual hand-eye servo	0.97	0.08	IEC 80601-2-77; ISO 9283; ASME V&V 40
2	Dexterous finger force	0.95	0.10	IEC 80601-2-77; ISO/TS 15066
3	Whole-body balance	0.98	0.06	ISO 13482; IEC 60601-1
4	Autonomous plan correctness	0.95	0.10	UL 4600; IEEE 7009; IEC 62304
5	Instructed grasp and handover	0.96	0.10	IEC 80601-2-77; ISO 9283
6	Vascular no-fly volume (hard)	1.00	0.05	IEC 80601-2-77; ISO 14971
7	Bimanual suturing anastomosis	0.96	0.08	IEC 80601-2-77; Fistula Risk Score
8	Perception scene understanding	0.95	0.10	ASME V&V 40; IEC 62304
9	Shared-room collaboration (hard)	1.00	0.06	ISO/TS 15066; ISO 10218-1; ISO 13482
10	Fault and emergency stop (hard)	1.00	0.05	IEC 60601-1; ISO 13849-1; IEEE 7009

Table 4: The ten-gate threshold schedule of section 515D(c): each gate’s minimum validation agreement, maximum coefficient of variation, and bound consensus standards. The three hard catastrophe predicates must hold at agreement 1.00.

Safety is answered by a mechanism, not a promise: a ten-gate floor bound to standards the relevant professions already trust, with three predicates that can stop the system cold, all required before the code is even generated.

5 Fiscal: what does it cost, and is it paid for?

The fourth question decides whether the bill can move at all. A member who cannot explain the score will not carry the bill, and a bill that trips a budget point of order does not reach the floor. H. R. 9510 is written to be scored cleanly.

The bill only authorizes; it does not appropriate, and it creates no entitlement. It authorizes appropriations to the FDA Salaries and Expenses account of 18 million dollars in fiscal year 2027, 12 million in 2028, 10 million in 2029, and 9 million in each of 2030 and 2031, for a five-year total of 58 million dollars, with estimated outlays of 53.5 million dollars inside the window and the small remainder after it [1]. Crucially for scorekeeping, the bill creates no direct spending and no revenue effect, so it carries no mandatory cost under the Statutory Pay-As-You-Go Act and adds nothing to the House CutGo scorecard [18, 19]. The only fee change is a carve-out: a verification record filed on its own is not a submission and bears no user fee, and the bill leaves the pricing of any incremental review workload to the next medical-device user-fee reauthorization rather than imposing a new statutory fee. Table 5 sets out the schedule, and a formal Congressional Budget Office estimate would follow introduction in the ordinary course [20].

Fiscal year	Authorization	Principal activity
FY 2027	\$18 million	Rulemaking under section 515D(h)
FY 2028	\$12 million	Verification-records system build
FY 2029	\$10 million	Steady-state operation and inspection
FY 2030	\$9 million	Steady-state operation
FY 2031	\$9 million	Steady-state, then transition to user fees
Total	\$58 million	\$53.5 million outlaid within the window; no direct spending

Table 5: The authorization schedule of section 5 and its budgetary effect: a bounded, declining authorization with no net mandatory spending.

The decisive number for a skeptical member is the cost asymmetry made concrete. A single cleared interaction class costs about 4,420 dollars to verify, of which the ten-gate compute is roughly 3,710 dollars, while the conventional path of manual code review and serial validation runs near 3 million dollars in a program-year; the automated gate operates the same program-year for about 155,000 dollars, a reduction of roughly nineteen to one [1, 21]. ?? places the gate’s small, predictable cost against the unbounded cost of the failure it prevents.

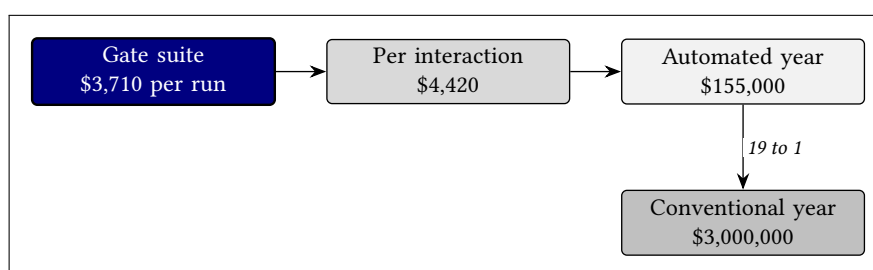


Figure 5. The cost asymmetry: a small, fixed verification cost against the unbounded cost it removes, a roughly nineteen-to-one reduction over conventional review; based on catalog diagrams 07 and 08.

Fiscal is answered with a clean score: a bounded authorization, no mandatory spending, no new fee, and a verification cost an order of magnitude below the status quo, which is precisely what a member needs to explain to the Budget Committee.

6 Constituents: who does this help back home?

The fifth question is local. A member votes for what helps the district and the State, and a national platform that reaches every region gives every member a concrete reason to be for the bill.

The platform that H. R. 9510 governs is designed to scale across the country. Its implementation strategy moves from a single site, to five through ten multi-state sites, to a steady state of 50 through 100 operational sites spanning all major geographic regions, with rolling activation of new sites each quarter [2, 22]. For a constituent, that scale translates into faster enrollment through automated screening, lower per-patient cost as fixed infrastructure is amortized over more patients, and access to advanced oncology trials closer to home, including in rural and underserved areas that rarely host a trial today. The convergence of human and machine performance that makes this possible is the same one documented across high-performance medicine [23]. ?? shows the rollout, and Table 6 lists what each phase delivers to a district.

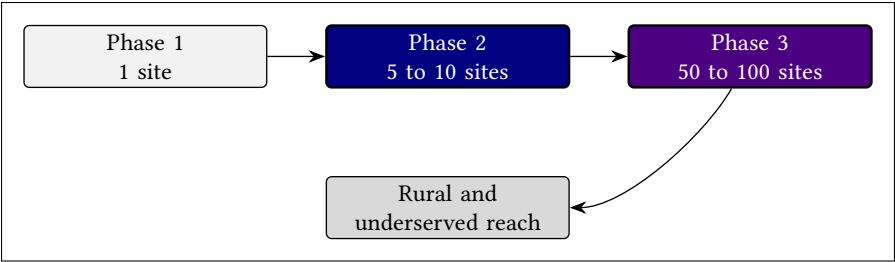


Figure 6. The national rollout from one site to a steady state of 50 through 100, reaching rural and underserved districts; the colored TikZ reproduction of catalog diagram 11.

What the district gets	How the platform delivers it
Faster trial access	Automated screening and eligibility checks compress the time from referral to enrollment.
Lower patient cost	Fixed infrastructure is amortized across more patients, so per-patient cost falls as volume rises.
Rural and underserved reach	Multi-site deployment targets all major regions rather than a few coastal centers.
New skilled jobs	Site operation creates roles in Physical AI operation, robot maintenance, and AI safety monitoring.

Table 6: Constituent benefit by phase: what the platform the bill enables delivers to a member’s district and State.

Constituents are answered with reach: a platform that, once the bill makes its safety floor law, can bring advanced oncology trials and skilled work to districts that have never had either.

7 Bipartisanship: can one caucus support it without losing the other?

The sixth question decides whether a bill is a message or a law. A purely partisan bill dies in the other chamber; H. R. 9510 is unusual because the same provision answers two caucuses that start from different premises.

To a member focused on innovation and a lighter regulatory hand, verification before generation is a predictable, automatable rule that lets credible systems deploy faster, replaces slow and inconsistent manual review with a fixed gate, and creates no new entitlement, tax, or fee. That posture is consistent with the executive direction to remove barriers to American leadership in artificial intelligence [24]. To a member focused on patient safety and human oversight, the same section is a hard safety floor: a ten-gate examination with three catastrophe predicates that cannot be waived, a preserved human-over-AI role, and a tamper-evident record an auditor can read. Neither side has to adopt the other’s reasoning to reach the same vote. ?? shows the convergence.

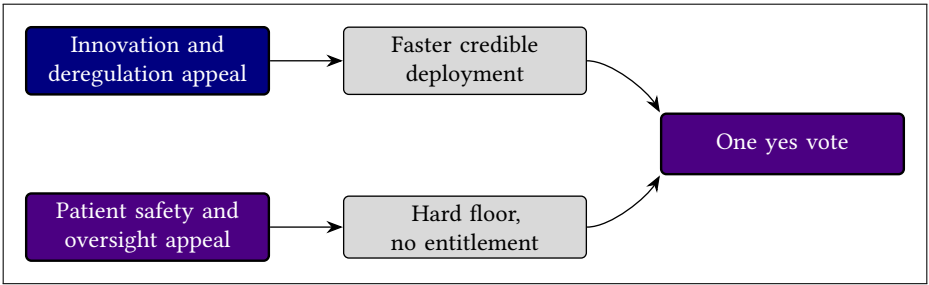


Figure 7. Two appeals, one vote: an innovation rationale and a safety rationale converge on the same provision; the colored TikZ reproduction of catalog diagram 12. Both curved arrows carry an explicit looseness of 0.7.

The political science of getting a bill across the aisle supports the approach. Legislators move when a case is framed for their own priorities and paired with credible evidence rather than advocacy alone, and the same evidence can be read by both sides when ambiguity, not just uncertainty, is reduced [25, 26, 27]. The framework's discipline of pairing every claim with a cited fact is built for exactly that.

Bipartisanship is answered by design: a single mechanism that a deregulation-minded member and a safety-minded member can each support on their own terms, with no daylight that an opponent can exploit.

8 Coalition: who is behind it, and is the opposition manageable?

The seventh question is about the room outside the chamber. A member counts who will testify for the bill, who will score it, and who, if anyone, will organize against it, because a bill with a broad coalition and no determined opponent is a bill a whip can move.

The coalition for H. R. 9510 assembles from groups that rarely align. Oncology and patient-safety organizations have a clear interest in a mandatory safety floor before an autonomous robot acts. Patient advocates and rural-access coalitions benefit from the platform's reach into underserved districts. Device and robotics manufacturers and the standards bodies gain predictable rules of the road, because the gates are bound to the consensus standards those bodies already publish [11, 13, 14]. The States keep their human-over-AI laws through the savings clause. The author's prior analysis of how proposed bills rank by patient priority places verification before generation at the top of the list a patient would choose [28]. ?? shows the coalition radiating from the bill.

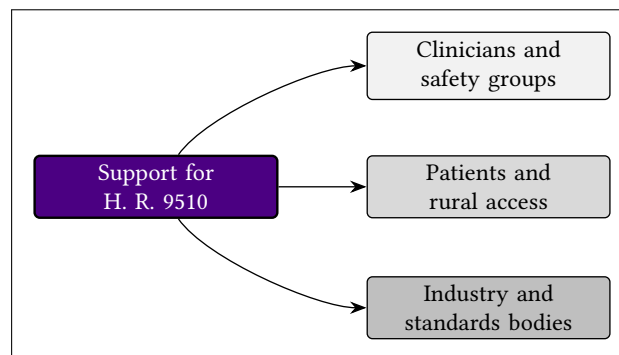


Figure 8. The supporting coalition radiates from the bill into clinicians, patients, and industry; the colored TikZ reproduction of catalog diagram 13. Each curved arrow carries an explicit looseness of 0.6.

The opposition is manageable because the bill gives little to organize against. It creates no entitlement to cut, imposes no new tax or fee, preempts little, and binds its rigor to standards the industry helped write. The predictable objections, that it costs too much, that it is federal overreach, that it will chill innovation, or that it is redundant, each have a specific answer in the text: the clean score, the savings clause, the change-control carve-outs, and the fact that the closest analogues remain voluntary. Routing each objection to its cited answer, rather than avoiding it, is what keeps a coalition together through markup.

Coalition is answered by breadth and the absence of a natural enemy: many groups gain, few lose, and every standard objection has a citation waiting.

9 Passage path: can it move through both chambers and then work?

The eighth question closes the loop. Even a good bill needs a path, and a member wants to know it can both pass and then be implemented without embarrassment.

The path is ordinary regular order, run in parallel across the chambers. The bill is introduced and referred to committee, refined in subcommittee and full-committee markup, and passed on the House floor; a Senate companion follows the same sequence; and the two versions are reconciled by amendment exchange or conference before the bill is enrolled and presented [29]. The thresholds are the familiar ones: a simple majority of the House, sixty votes to invoke cloture in the Senate, and a simple majority to pass it. The appendix figures trace this journey, the two-chamber exchange, the markup as a branch-and-merge of the text, and the vote thresholds as formal requirements.

Implementation is already scoped, which is what lets a member promise the bill will work. Section 515D directs the Secretary to issue final regulations within 365 days, takes effect on the first day of the first calendar quarter at least one year after enactment, and gives existing enrolling investigations 180 days to comply, so the runway is defined rather than open-ended [1]. More important, the system the bill governs has already been demonstrated. In validated simulation the National Platform treated 168 patients with 29 robots across 15 cancer types at 99.7 percent uptime with zero patient harm events, and it is built to operate as a fully automated sponsor across many sites [2, 30]. ?? closes the framework by tracing one member's yes up to the statute and the platform ready to implement it.

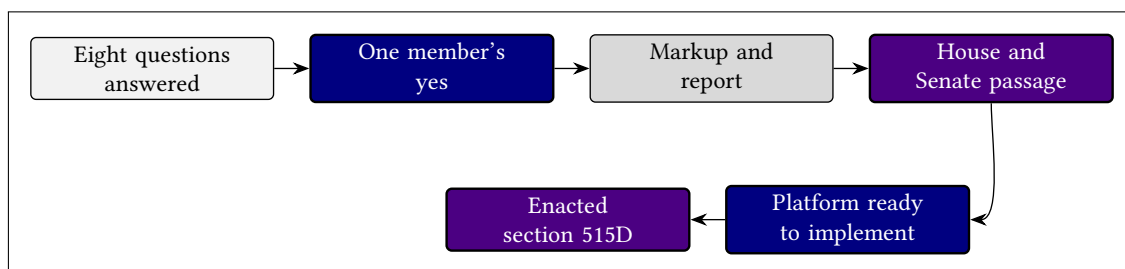


Figure 9. From one member's yes to enacted law and a platform ready to implement it; the colored TikZ reproduction of catalog diagram 20.

The passage path is answered with a plan: ordinary regular order in both chambers, known vote thresholds, a defined implementation runway, and a platform already shown to work, so a yes vote is a vote for something that will function.

10 Appendix A: figure compendium

This appendix reproduces the remaining figures from the Mermaid catalog (`enactment/mermaid/`) as colored TikZ, so the framework carries the full set of twenty exhibits in the strict palette (black, grayscales, #4B0082, #000080, #C0C0C0). Every figure keeps a uniform `\vspace{-0.75cm}` between the figure and its caption.

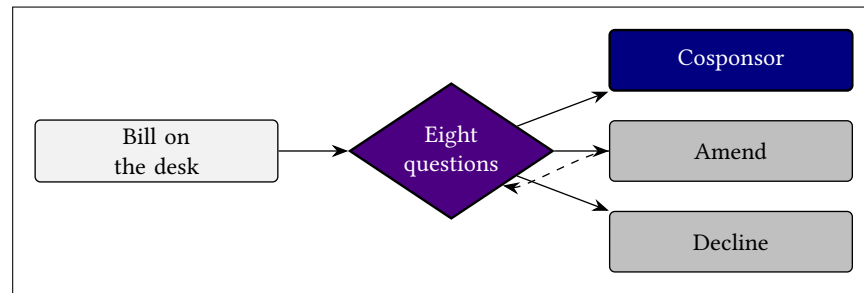


Figure 10. The passage decision: the eight questions gate one choice with three guarded outcomes, and an amendment adopted at markup returns to the questions (diagram 01).

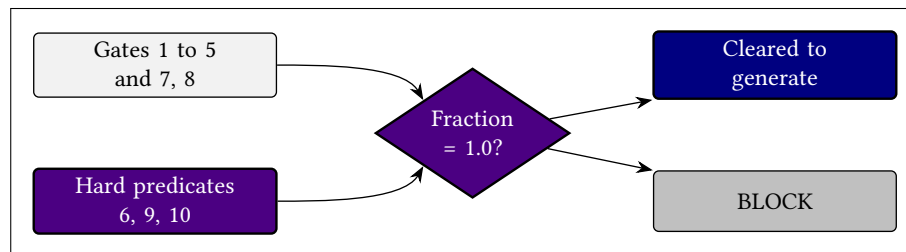


Figure 11. The ten gates Congress is enacting: the routine gates and the three hard catastrophe predicates both feed the verification-fraction test (diagram 04).

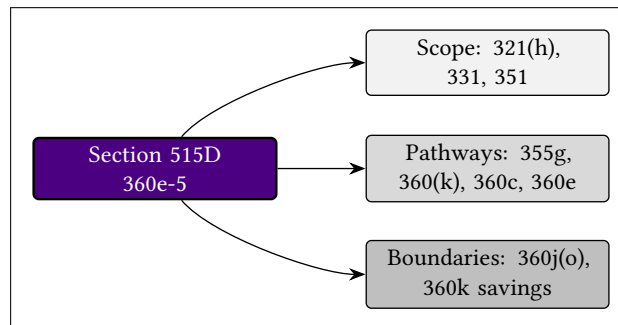


Figure 12. The amendment map: one new section radiates ten conforming amendments into Title 21 (diagram 05).

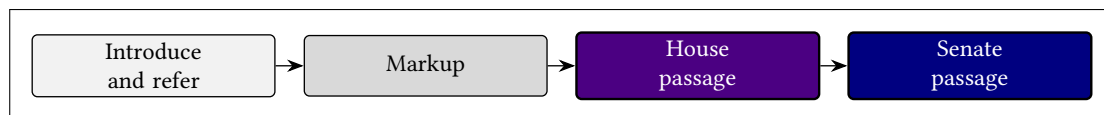


Figure 13. The legislative journey from introduction through House and Senate passage (diagram 06).

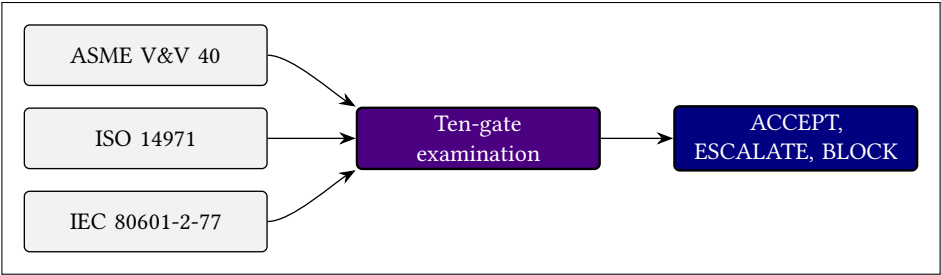


Figure 14. The recognized standards basis: consensus standards flow into the one examination the statute requires (diagram 10).

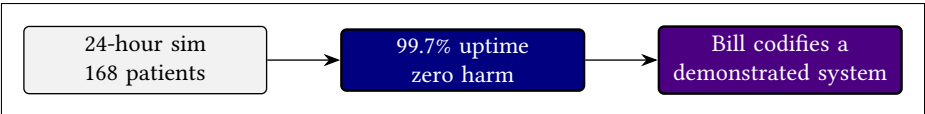


Figure 15. The platform as proof of implementability: validated simulation evidence supports codification (diagram 14).

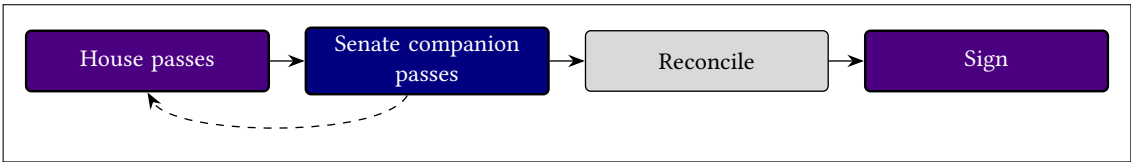


Figure 16. The two-chamber strategy: a Senate companion runs in parallel and the versions are reconciled before signature (diagram 15).

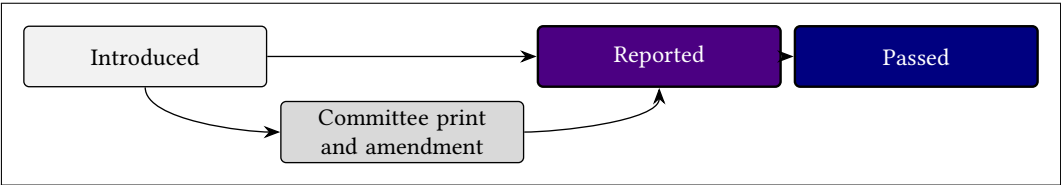


Figure 17. The markup lifecycle as a branch and merge of the text: a committee branch refines the bill before it merges back as reported (diagram 16).

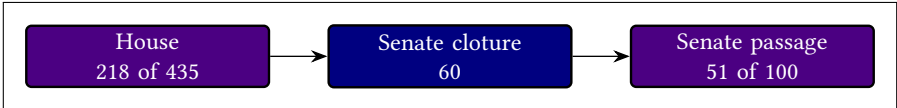


Figure 18. The vote thresholds passage must satisfy in each chamber (diagram 17).

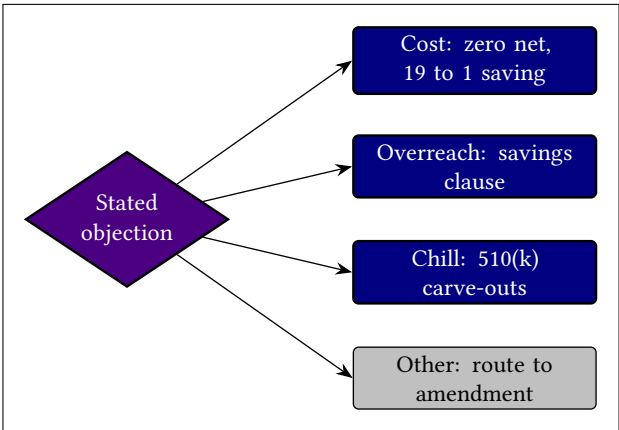


Figure 19. Handling the standard objections: each predictable objection routes to a cited answer, and only a new concern routes to an amendment (diagram 18).

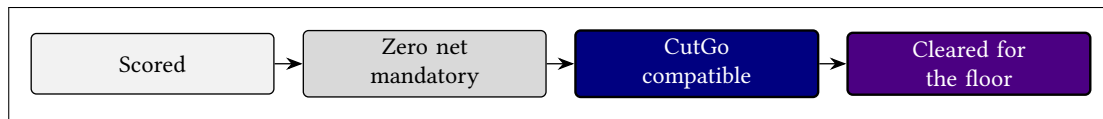


Figure 20. *The scoring path: a discretionary authorization clears the budget points of order and reaches the floor (diagram 19).*

Taken together, the twenty figures trace one claim from a single member's reading of the bill (Figure 10) to enactment in both chambers and a platform ready to implement it (Figure 9). None asks a member to vote on faith; each shows where a gate, a score line, a provision, or a coalition makes the yes defensible.

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Ethical disclosures

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Data availability

All source files, the Mermaid catalog, and the references are available in the *enactment/* directory of the *kevinkawchak/law-physical-ai-trials* repository. No proprietary data is used; all supporting numbers are illustrative or simulation results unless tied to a cited source.